State-Weighted Defect Transfer for Dependent Block Explorations

Abstract

We prove a weighted positive-kernel criterion for exponential suppression of connected bad-block events. The criterion remembers energetic cost stored in a retained boundary state and can remain subcritical even when a scalar worst-history conditional probability equals one. It is an abstract finite transfer theorem; construction of the required Yang–Mills kernel remains an analytic obligation.

1 Weighted exploration

Let Σ be a measurable retained-state space, $V : \Sigma \to [1, \infty)$, and let $Q_i(s, ds')$ be the subprobability kernel for a bad extension at step i. Assume uniformly in i and s that

$$\int V(s')Q_i(s, ds') \le qV(s). \tag{1}$$

Theorem 1 (Weighted chain bound)**.** *If* (1) *holds and the initial law* ν *satisfies* $\nu(V) \le M$*, then any specified chain of* n *bad extensions has probability at most* Mq^n*.*

Proof. Positivity and repeated kernel composition give

$$\int \nu(ds_0)Q_1(s_0, ds_1) \cdots Q_n(s_{n-1}, ds_n)V(s_n) \le q^n\nu(V) \le Mq^n.$$

Since $V(s_n) \ge 1$, deleting the terminal weight can only decrease the integrand on the bad-chain event. □

Theorem 2 (Rooted connected-set sum)**.** *On a graph of maximum degree* Δ*, suppose every connected bad set can be exposed by successive adjacent bad extensions satisfying the preceding hypothesis. If* $e^\alpha\Delta^2 q < 1$*, then*

$$\sum_{\substack{X \ni o \\ X \text{ connected}}} e^{\alpha|X|}\mathbb{P}(B_X) \le \frac{Me^\alpha q}{1 - e^\alpha\Delta^2 q}.$$

Proof. A rooted connected set of size n admits a depth-first traversal of length at most $2(n-1)$, giving the standard upper bound $\Delta^{2(n-1)}$ on its number. Sum the weighted chain estimate over n. □

2 Why the state weight matters

Example 1. *For* $0 < q < 1$*, take*

$$Q = \begin{pmatrix} 0 & q^2 \\ 1 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} 1 \\ q^{-1} \end{pmatrix}.$$

The scalar row ceiling is one, but $QV = qV$. Entry into the high-cost state is rare, although the following bad transition is forced. For finite irreducible Q, minimizing $\max_s (QV)(s)/V(s)$ over positive V yields the Perron–Frobenius spectral radius.

At $q = 0.002430277501841998$, $\Delta = 18$, and $\alpha = 0.1$, the branching ratio is 0.8702225338 and the rooted sum is 0.0206959813. The strict critical value is

$$q_{\mathrm{crit}} = e^{-0.1}/18^2 = 0.0027927080803578997,$$

equivalently a weighted extension cost greater than 5.8807435158.

3 Scope

The theorem does not identify Q with a nonlinear lattice Yang–Mills conditional kernel. The next obligation is to construct a gauge-covariant retained state and physical weight satisfying the strict ceiling uniformly in volume. No RG, continuum, or Hamiltonian-gap claim follows at this stage.